

Superstore Performance & Profitability Analysis

BUSINESS INTELLIGENCE, DAX ENGINEERING & EXECUTIVE PERFORMANCE DASHBOARD

PROJECT OVERVIEW

This project delivers a data-driven evaluation of sales, profitability, and operational efficiency for a global retail business. By analyzing historical transaction data across thousands of orders, the objective was to design and engineer an end-to-end Power BI reporting ecosystem. The resulting dashboard uncovers critical margin leakage points, identifies high-performing regional hubs, and segments customer behavior profiles to guide executive retail strategy.

DATASET SUMMARY

- **Total Transactions:** 9,994 retail order lines
- **Unique Orders:** 5,009 processed orders
- **Active Customer Base:** 793 unique buyers
- **Key Attributes Analyzed:** Temporal tracks (Order & Ship Dates), Shipping Logistics, Customer Profiles, Geographic Variables (Regions, States, Cities), Product Hierarchies (Categories & Sub-Categories), and Core Financial Metrics (Sales, Quantity, Discounts, Profit).

DATA MODELING & DAX ENGINEERING

The entire architecture was built inside Power BI Desktop, utilizing Power Query for structural adjustments and establishing an analytical schema to support deep-dive visual profiling:

- **Data Profiling & Transformation:** Cleaned data columns inside Power Query. Formatted date parameters, verified geographical data categories (State, City, Country) for accurate spatial mapping, and structured numeric types for Sales and Profit fields.
- **Relational Schema Setup:** Formulated a clean relational model inside the Model View. Created a Single-to-Many relationship map connecting data nodes to isolate customer activities across individual transaction lines.
- **Advanced Measure Formulations:** Written customized Data Analysis Expressions (DAX) to go beyond standard aggregations and power the visualization layer:

```
Total Revenue = SUM('Sample - Superstore'[Sales])
Net Profit = SUM('Sample - Superstore'[Profit])
```

```
Profit Margin % = DIVIDE([Net Profit], [Total Revenue], 0)
Total Orders = DISTINCTCOUNT('Sample - Superstore'[Order ID])
```

- **Dynamic Behavioral Cohort Logic:** Engineered a dynamic DAX Calculated Table to group the user base into distinct customer lifecycle segments based on purchase recency, order frequencies, and monetary volumes:

```
CustomerSegments =
VAR MaxDate = MAX('Sample - Superstore'[Order Date])
VAR Summary = SUMMARIZE('Sample - Superstore', 'Sample - Superstore'[Customer ID], 'Sample - Superstore'[Customer Name])
VAR Metrics = ADDCOLUMNS(Summary, "SalesSum", CALCULATE(SUM('Sample - Superstore'[Sales])),
"OrdersCount", CALCULATE(DISTINCTCOUNT('Sample - Superstore'[Order ID])), "LastDate",
CALCULATE(MAX('Sample - Superstore'[Order Date])))
VAR DaysOut = ADDCOLUMNS(Metrics, "DaysSinceLast", VALUE(MaxDate - [LastDate]))
RETURN
    ADDCOLUMNS(DaysOut, "Customer Segment",
        SWITCH(TRUE(),
            [DaysSinceLast] > 365, "Lapsed / Churned",
            [DaysSinceLast] > 180, "Churn-Risk",
            [OrdersCount] >= 8 || [SalesSum] >= 3500, "High-Value / Champions",
            [OrdersCount] >= 4, "Loyal / Regular",
            "Occasional / New"
        )
    )
```

EXECUTIVE BI DASHBOARD ARCHITECTURE

The visual layout was built with strict information architecture principles to translate complex retail data tables into immediate corporate strategy:

- **KPI Control Strip:** Placed high-level analytical cards at the top of the viewport to display essential health indicators: Total Sales (\$2.3M), Net Profit (\$286.4K), Unique Transactions (5,009), and Active Customers (793).
- **Geographic Profit Matrix:** Integrated spatial visual mapping paired with conditional color formatting to contrast gross volume against net profit margins. This instantly flags underperforming sales territories.
- **Customer Health Breakdown:** Deployed an executive Donut Chart visualization displaying the distribution of unique accounts across the newly engineered behavioral cohorts.
- **Cross-Visual Slicing Filters:** Configured interactive dashboard parameters allowing users to dynamically filter performance maps, product ranks, and operational timelines by Customer Segments or time periods.



Figure 1: Finished Power BI Executive Analytics Interface showcasing core KPI strips, operational timelines, and category breakdowns.

CORE BUSINESS INSIGHTS & REVELATIONS

1. The Discounting Paradox & Profit Leakage

- **The Threshold of Loss:** Cross-filtering financial measures against discount rates reveals a major structural issue: product promotions exceeding 40% lead to severe profit degradation rather than self-sustaining sales volumes.
- **Product Line Inefficiencies:** Product segment analysis uncovers that **Tables (-\$17,725)** and **Bookcases (-\$3,473)** act as significant financial drains, consistently losing money despite capturing major sales volume.

2. Regional Performance Disparities

- **Profitable Engines vs. Deficit Hubs:** The West and East regions remain highly stable and profitable. However, major sales states like **Texas (-\$25,729)**, **Ohio (-\$16,971)**, and **Pennsylvania (-\$15,560)** suffer severe profit deficits. This highlights localized margin erosion, likely driven by regional price matching and shipping overheads.

3. Behavioral Cohort Dynamics

- **Retention Value Over Acquisition:** The user segmentation model confirms that a small, highly engaged cohort of **High-Value Champions** generates a disproportionate share of net margin stability.

- **Retention Alarms:** The dashboard captures a substantial number of historical customer profiles slipping into the 180-to-365-day inactivity zone (Churn-Risk), signaling immediate operational attrition before permanent customer loss occurs.

STRATEGIC RECOMMENDATIONS

- **Implement Automated Discount Ceilings:** Put a hard 20% limit on standard retail furniture promotions in Power BI's underlying policy data. Any discount above this threshold should require corporate overrides or be reserved for bulk corporate contracts.
- **Restructure Local Pricing Tiers:** Move away from flat national pricing models. Adjust pricing tiers in deficit states (Texas, Ohio) within the system to automatically factor in localized delivery, freight, and freight-handling overhead.
- **Product Bundling Strategy:** Create high-margin storefront product bundles that tie underperforming, loss-leading assets (Tables, Bookcases) directly to high-margin accessories (Technology peripherals, high-retention Office Supplies).
- **Data-Driven Win-Back Campaigns:** Use the Customer Segment classifications to target users in the "Churn-Risk" zone with tailored re-engagement incentives before they shift to the "Lapsed" status.